

If Else Questions

1. Largest Among Three Numbers

Write a C program to input 3 numbers and print the largest number using if-else.

Program:

```
#include <stdio.h>

int main()
{
    int num1, num2, num3, large;
    printf("Enter 3 numbers: ");
    scanf("%d%d%d", &num1, &num2, &num3);
    if(num1 > num2 && num1 > num3)
    {
        large = num1;
    }
    else if(num2 > num3)
    {
        large = num2;
    }
    else
    {
        large = num3;
    }
    printf("Largest number is: %d", large);
    return 0;
}
```

Output:

```
Enter 3 numbers: 50 100 62
Largest number is: 100
```

2. Leap Year Checker

Write a program to check whether a given year is a leap year or not.

Rules:

- Divisible by 4
- But not divisible by 100
- Unless divisible by 400

Program:

```
#include <stdio.h>

int main()
{
    int year;
    printf("Enter the year: ");
```

```

scanf("%d", &year);
if((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0))
{
    printf("Entered year is a leap year");
}
else
{
    printf("Entered year is not a leap year");
}
return 0;
}

```

Output:

```

Enter the year : 2028
Entered year is leap year

```

3. Electricity Bill Generator

Create a program to calculate electricity bill based on units:

Units	Rate
0-100	Rs.5/unit
101-200	Rs.7/unit
201+	Rs.10/unit

Add 18% GST at the end.

Program:

```

#include <stdio.h>

int main() {
    float units;
    float slab1_cost = 0, slab2_cost = 0, slab3_cost = 0;
    float base_amount, gst_amount, total_bill;
    printf("Enter the total units consumed: ");
    if (scanf("%f", &units) != 1) {
        printf("Invalid input. Please enter a valid number.\n");
        return 1;
    }
    if (units < 0) {
        printf("Units consumed cannot be negative.\n");
        return 1;
    }
    if (units <= 100) {
        slab1_cost = units * 5;
    } else if (units <= 200) {
        slab1_cost = 100 * 5;
        slab2_cost = (units - 100) * 7;
    }
}

```

```

    } else {
        slab1_cost = 100 * 5;
        slab2_cost = 100 * 7;
        slab3_cost = (units - 200) * 10;
    }
    base_amount = slab1_cost + slab2_cost + slab3_cost;
    gst_amount = base_amount * 0.18;
    total_bill = base_amount + gst_amount;
    printf("\n=====\\n");
    printf("          ELECTRICITY BILL\\n");
    printf("=====\\n");
    printf("Total Units Consumed: %.2f\\n", units);
    printf("-----\\n");
    printf("Slab 1 (0-100 units) : Rs. %.2f\\n", slab1_cost);
    printf("Slab 2 (101-200 units): Rs. %.2f\\n", slab2_cost);
    printf("Slab 3 (201+ units) : Rs. %.2f\\n", slab3_cost);
    printf("-----\\n");
    printf("Base Amount          : Rs. %.2f\\n", base_amount);
    printf("GST (18%)            : Rs. %.2f\\n", gst_amount);
    printf("=====\\n");
    printf("TOTAL PAYABLE AMOUNT : Rs. %.2f\\n", total_bill);
    printf("=====\\n");
    return 0;
}

```

Output:

Enter the total units consumed: 200

```

=====
          ELECTRICITY BILL
=====
Total Units Consumed: 200.00
-----
Slab 1 (0-100 units) : Rs. 500.00
Slab 2 (101-200 units): Rs. 700.00
Slab 3 (201+ units) : Rs. 0.00
-----
Base Amount          : Rs. 1200.00
GST (18%)            : Rs. 216.00
=====
TOTAL PAYABLE AMOUNT : Rs. 1416.00
=====

```

4. Grade System

Input marks of 5 subjects and calculate percentage.

Print:

- A Grade -> 90+
- B Grade -> 75+
- C Grade -> 50+
- Fail -> below 50

Also print: Distinction / Pass / Fail

Program:

```
#include <stdio.h>

int main() {
    float m1,m2,m3,m4,m5,total,percentage;
    printf("Enter marks of 1st subject : ");
    scanf("%f",&m1);
    printf("Enter marks of 2nd subject : ");
    scanf("%f",&m2);
    printf("Enter marks of 3rd subject : ");
    scanf("%f",&m3);
    printf("Enter marks of 4th subject : ");
    scanf("%f",&m4);
    printf("Enter marks of 5th subject : ");
    scanf("%f",&m5);

    total=m1+m2+m3+m4+m5;
    percentage=total*100/500;
    printf("percentage = %.2f\n\n",percentage);

    if(percentage >= 90)
    {
        printf("Grade: A\n");
        printf("Result: Distinction");
    }
    else if(percentage >= 75)
    {
        printf("Grade: B\n");
        printf("Result: First Class");
    }
    else if(percentage >= 50)
    {
        printf("Grade: C\n");
        printf("Result: Pass");
    }
    else
    {
        printf("Grade: Fail\n");
        printf("Result: Fail");
    }
    return 0;
}
```

Output:

```
Enter marks of 1st subject : 95
Enter marks of 2nd subject : 92
Enter marks of 3rd subject : 97
Enter marks of 4th subject : 94
Enter marks of 5th subject : 90
percentage = 93.60
Grade: A
Result: Distinction
```

5. ATM Withdrawal System

Input:

- Account balance
- Withdrawal amount

Conditions:

- Minimum balance should remain Rs.1000
- Withdrawal should be multiple of 100
- Print success/failure message

Program:

```
#include <stdio.h>

int main()
{
    int balance, withdraw;
    printf("Enter account balance: ");
    scanf("%d", &balance);
    printf("Enter withdrawal amount: ");
    scanf("%d", &withdraw);
    if(withdraw % 100 != 0)
    {
        printf("Withdrawal amount should be multiple of 100");
    }
    else if(balance - withdraw < 1000)
    {
        printf("Transaction failed! Minimum balance of Rs.1000 must remain");
    }
    else
    {
        balance = balance - withdraw;
        printf("Withdrawal successful!\n");
        printf("Remaining balance = Rs.%d", balance);
    }
    return 0;
}
```

Output:

```
Enter account balance: 1500
Enter withdrawal amount: 600
Transaction failed! Minimum balance of Rs.1000 must remain
```

6. Login Authentication System

Create a simple login system.

Correct credentials:

- Username: admin
- Password: 1234

Use nested if-else.

Program:

```
#include <stdio.h>
#include <string.h>

int main() {
    char username[5];
    int password;
    printf("Enter Username : ");
    scanf("%s", username);
    printf("Enter Password : ");
    scanf("%d", &password);

    if (strcmp(username, "admin") == 0)
    {
        if (password == 1234)
        {
            printf("Login Successful");
        }
        else
        {
            printf("Incorrect Password");
        }
    }
    else
    {
        printf("Incorrect Username");
    }
    return 0;
}
```

Output:

```
Enter Username : admin
Enter Password : 1234
Login Successful
```

7. Triangle Validator

Input 3 angles.

Check:

- Whether triangle is valid
- Then print:
 - Equilateral
 - Isosceles
 - Scalene

Program:

```
#include <stdio.h>

int main()
{
    int a, b, c;
```

```

printf("Enter 3 angles of triangle: ");
scanf("%d%d%d", &a, &b, &c);
if(a + b + c == 180)
{
    printf("Triangle is valid\n");
    if(a == b && b == c)
    {
        printf("Equilateral Triangle");
    }
    else if(a == b || b == c || a == c)
    {
        printf("Isosceles Triangle");
    }
    else
    {
        printf("Scalene Triangle");
    }
}
else
{
    printf("Triangle is not valid");
}
return 0;
}

```

Output:

```

Enter 3 angles of triangle: 90 50 40
Triangle is valid
Scalene Triangle

```

8. Employee Salary Bonus System

Input:

- Salary
- Years of experience
- Performance rating

Conditions:

- Experience > 5 years -> 20% bonus
- Rating A -> Extra 10%
- Rating B -> Extra 5%
- Otherwise no extra bonus

Print final salary.

Program:

```

#include <stdio.h>

int main()
{
    float salary, bonus = 0, finalSalary;
    int experience;

```

```

    char rating;
    printf("Enter salary: ");
    scanf("%f", &salary);
    printf("Enter years of experience: ");
    scanf("%d", &experience);
    printf("Enter performance rating (A/B/C): ");
    scanf(" %c", &rating);
    if(experience > 5)
    {
        bonus = bonus + (salary * 0.20);
    }
    if(rating == 'A')
    {
        bonus = bonus + (salary * 0.10);
    }
    else if(rating == 'B')
    {
        bonus = bonus + (salary * 0.05);
    }
    finalSalary = salary + bonus;
    printf("Final Salary = Rs.%.2f", finalSalary);
    return 0;
}

```

Output:

```

Enter salary: 100000
Enter years of experience: 6
Enter performance rating (A/B/C): A
Final Salary = Rs.130000.00

```

9. E-Commerce Discount System

Create a shopping bill program.

Conditions:

- Above Rs.5000 -> 20% discount
- Above Rs.2000 -> 10% discount
- VIP customer -> extra 5%

Add GST after discount.

Program:

```

#include <stdio.h>

int main()
{
    float amount, discount = 0, gst, finalBill;
    char vip;
    printf("Enter shopping amount: ");
    scanf("%f", &amount);
    printf("Are you a VIP customer? (Y/N): ");
    scanf(" %c", &vip);
    if(amount > 5000)
    {

```

```

        discount = discount + (amount * 0.20);
    }
    else if(amount > 2000)
    {
        discount = discount + (amount * 0.10);
    }
    if(vip == 'Y' || vip == 'y')
    {
        discount = discount + (amount * 0.05);
    }
    amount = amount - discount;
    gst = amount * 0.18;
    finalBill = amount + gst;
    printf("Final Bill = Rs.%.2f", finalBill);
    return 0;
}

```

Output:

```

Enter shopping amount: 6500
Are you a VIP customer? (Y/N): y
Final Bill = Rs.5752.50

```

10. Online Voting System

Input:

- Age
- Citizenship (Indian or not)
- Criminal record (Yes/No)

Eligible only if:

- Age ≥ 18
- Indian citizen
- No criminal record

Program:

```

#include <stdio.h>

int main()
{
    int age;
    char Citizenship, Criminal_record;

    printf("Enter your age : ");
    scanf("%d", &age);
    printf("Are you Indian (Y/N) : ");
    scanf(" %c", &Citizenship);
    printf("Do you have any criminal record (Y/N) : ");
    scanf(" %c", &Criminal_record);

    if(age >= 18)
    {
        if(Citizenship == 'Y' || Citizenship == 'y')

```

```

        {
            if(Criminal_record=='N' || Criminal_record=='n')
            {
                printf("You are Eligible to vote");
            }
            else
            {
                printf("You are not Eligible to vote: having Criminal
record");
            }
        }
        else
        {
            printf("You are not Eligible to vote: Citizenship is not India");
        }
    }
    else
    {
        printf("You are not Eligible to vote: Age is less than 18");
    }
    return 0;
}

```

Output:

```

Enter your age : 20
Are you Indian (Y/N) : Y
Do you have any criminal record (Y/N) : N
You are Eligible to vote

```

11. Smart Traffic Signal System

Input vehicle speed.

Conditions:

- Speed < 40 -> Safe
- 40-80 -> Normal
- 81-120 -> Warning
- Above 120 -> Challan + License Suspension

Program:

```

#include <stdio.h>

int main()
{
    int speed;
    printf("Enter vehicle speed: ");
    scanf("%d", &speed);
    if(speed < 40)
    {
        printf("Safe Speed");
    }
    else if(speed <= 80)
    {

```

```

        printf("Normal Speed");
    }
    else if(speed <= 120)
    {
        printf("Warning! Overspeeding");
    }
    else
    {
        printf("Challan + License Suspension");
    }
    return 0;
}

```

Output:

```

Enter vehicle speed: 70
Normal Speed

```

12. Income Tax Calculator

Calculate tax based on income slabs:

Income	Tax
Up to Rs.2.5L	0%
Rs.2.5L - Rs.5L	5%
Rs.5L - Rs.10L	20%
Above Rs.10L	30%

Add 4% cess.

Program:

```

#include <stdio.h>

int main()
{
    float income, slab1 = 0, slab2 = 0, slab3 = 0;
    float total = 0, cess;
    printf("Enter your income: ");
    scanf("%f", &income);
    if(income <= 250000)
    {
        total = 0;
    }
    else if(income <= 500000)
    {
        slab1 = (income - 250000) * 0.05;
        total = slab1;
    }
}

```

```

    }
    else if(income <= 1000000)
    {
        slab1 = 250000 * 0.05;
        slab2 = (income - 500000) * 0.20;
        total = slab1 + slab2;
    }
    else
    {
        slab1 = 250000 * 0.05;
        slab2 = 500000 * 0.20;
        slab3 = (income - 1000000) * 0.30;
        total = slab1 + slab2 + slab3;
    }
    cess = total * 0.04;
    total = total + cess;
    printf("Total Tax = %.2f", total);
    return 0;
}

```

Output:

```

Enter your income: 700000
Total Tax = 54600.00

```

13. Railway Ticket Booking

Input:

- Age
- Gender
- Ticket class

Conditions:

- Senior citizen discount
- Female discount
- AC class extra charge

Print final fare.

Program:

```

#include <stdio.h>

int main()
{
    int age;
    char gender, ticketClass;
    float fare = 500, discount = 0, finalFare;
    printf("Enter age: ");
    scanf("%d", &age);
    printf("Enter gender (M/F): ");
    scanf(" %c", &gender);
    printf("Enter ticket class (A for AC / N for Normal): ");
    scanf(" %c", &ticketClass);
    if(age >= 60)

```

```

    {
        discount = discount + (fare * 0.20);
    }
    if(gender == 'F' || gender == 'f')
    {
        discount = discount + (fare * 0.10);
    }
    fare = fare - discount;
    if(ticketClass == 'A' || ticketClass == 'a')
    {
        fare = fare + 200;
    }
    finalFare = fare;
    printf("Final Fare = Rs.%.2f", finalFare);
    return 0;
}

```

Output:

```

Enter age: 70
Enter gender (M/F): f
Enter ticket class (A for AC / N for Normal): a
Final Fare = Rs.550.00

```

14. Student Result Management System

Input marks of:

- C
- Java
- DBMS
- HTML
- React

Conditions:

- Any subject below 35 -> Fail
- Otherwise calculate:
 - Percentage
 - Grade
 - Topper status (above 90%)

Use nested conditions.

Program:

```

#include <stdio.h>

int main()
{
    float c, java, dbms, html, react;
    float total, percentage;
    printf("Enter marks of C: ");
    scanf("%f", &c);
    printf("Enter marks of Java: ");

```

```

scanf("%f", &java);
printf("Enter marks of DBMS: ");
scanf("%f", &dbms);
printf("Enter marks of HTML: ");
scanf("%f", &html);
printf("Enter marks of React: ");
scanf("%f", &react);
if(c < 35 || java < 35 || dbms < 35 || html < 35 || react < 35)
{
    printf("Result: Fail");
}
else
{
    total = c + java + dbms + html + react;
    percentage = total / 5;
    printf("Percentage = %.2f\n", percentage);
    if(percentage >= 90)
    {
        printf("Grade: A\n");
        printf("Topper Status: Yes");
    }
    else if(percentage >= 75)
    {
        printf("Grade: B\n");
        printf("Topper Status: No");
    }
    else if(percentage >= 50)
    {
        printf("Grade: C\n");
        printf("Topper Status: No");
    }
    else
    {
        printf("Grade: Fail");
    }
}
return 0;
}

```

Output:

```

Enter marks of C: 95
Enter marks of Java: 94
Enter marks of DBMS: 90
Enter marks of HTML: 91
Enter marks of React: 96
Percentage = 93.20
Grade: A
Topper Status: Yes

```

15. Bank Loan Eligibility System

Input:

- Salary
- CIBIL score

- Existing loan
- Age

Conditions:

- Salary > Rs.25,000
- CIBIL > 750
- Existing loan < Rs.5L
- Age between 21-60

Print:

- Eligible
- Not Eligible
- Premium Customer Loan Offer

Program:

```
#include <stdio.h>

int main()
{
    float salary, existingLoan;
    int cibil, age;
    printf("Enter salary: ");
    scanf("%f", &salary);
    printf("Enter CIBIL score: ");
    scanf("%d", &cibil);
    printf("Enter existing loan amount: ");
    scanf("%f", &existingLoan);
    printf("Enter age: ");
    scanf("%d", &age);
    if(salary > 25000 && cibil > 750 && existingLoan < 500000 && age >= 21 &&
age <= 60)
    {
        printf("Eligible for Loan\n");
        if(salary > 100000 && cibil > 800)
        {
            printf("Premium Customer Loan Offer");
        }
    }
    else
    {
        printf("Not Eligible for Loan");
    }
    return 0;
}
```

Output:

```
Enter salary: 1500000
Enter CIBIL score: 900
Enter existing loan amount: 250000
Enter age: 22
Eligible for Loan
Premium Customer Loan Offer
```

For Loop Questions

1. Factorial of Number

Calculate factorial using for loop.

Example: $5! = 120$

Program:

```
#include<stdio.h>

int main()
{
    int num, fact=1;
    printf("Enter the number : ");
    scanf("%d",&num);

    for(int i=num; i>0; i--)
    {
        fact=fact*i;
    }
    printf("Factorial of %d is %d", num, fact);
    return 0;
}
```

Output:

```
Enter the number : 5
Factorial of 5 is 120
```

2. Reverse Table

Print multiplication table in reverse order.

Example: $5 \times 10 = 50$, $5 \times 9 = 45$...

Program:

```
#include<stdio.h>

int main()
{
    int num;
    printf("Enter any number : ");
    scanf("%d",&num);

    for(int i=10; i>=1; i--)
    {
        printf("%d * %d = %d\n", num, i, (num*i));
    }
    return 0;
}
```

Output:

```
Enter any number : 5
5 * 10 = 50
5 * 9 = 45
5 * 8 = 40
5 * 7 = 35
5 * 6 = 30
5 * 5 = 25
5 * 4 = 20
5 * 3 = 15
5 * 2 = 10
5 * 1 = 5
```

3. Count Digits

Input a number and count total digits using loop.

Program:

```
#include<stdio.h>

int main()
{
    int num, count=0;
    printf("Enter any number : ");
    scanf("%d", &num);

    while(num!=0)
    {
        num=num/10;
        count++;
    }
    printf("Total no of digits in the given no is : %d", count);
    return 0;
}
```

Output:

```
Enter any number : 1234
Total no of digits in the given no is : 4
```

4. Sum of Digits

Find sum of digits of a number.

Example: 1234 -> 10

Program:

```
#include<stdio.h>
```

```
int main()
{
    int num, sum=0, digit;
    printf("Enter any number : ");
    scanf("%d", &num);

    while(num!=0)
    {
        digit=num%10;
        sum=sum+digit;
        num=num/10;
    }
    printf("Sum of digits in the given no is : %d", sum);
    return 0;
}
```

Output:

```
Enter any number : 1234
Sum of digits in the given no is : 10
```

5. Reverse Number

Reverse a number using loop.

Example: 1234 -> 4321

Program:

```
#include<stdio.h>

int main()
{
    int num, rev=0, digit;
    printf("Enter any number : ");
    scanf("%d", &num);

    while(num!=0)
    {
        digit=num%10;
        rev=(rev*10)+digit;
        num=num/10;
    }
    printf("Reverse no is : %d", rev);
    return 0;
}
```

Output:

```
Enter any number : 1234
Reverse no is : 4321
```

6. Strong Number

Check whether number is a Strong Number.

Example: 145 -> Strong Number

Program:

```
#include<stdio.h>

int main()
{
    int num, digit, fact, temp, sum=0;
    printf("Enter any number : ");
    scanf("%d",&num);
    temp=num;

    while(num!=0)
    {
        digit=num%10;
        fact=1;
        for(int i=digit; i>=1; i--)
        {
            fact=fact*i;
        }
        sum=sum+fact;
        num=num/10;
    }
    if(sum == temp)
    {
        printf("Strong Number");
    }
    else
    {
        printf("Not a Strong Number");
    }
    return 0;
}
```

Output:

```
Enter any number : 145
Strong Number
```

7. Find HCF and LCM

Find HCF and LCM of two numbers using loops.

Program:

```
#include <stdio.h>

int main()
{
    int num1, num2, hcf, lcm, i;
    printf("Enter two numbers: ");
```

```

scanf("%d%d", &num1, &num2);
for(i = 1; i <= num1 && i <= num2; i++)
{
    if(num1 % i == 0 && num2 % i == 0)
    {
        hcf = i;
    }
}
lcm = (num1 * num2) / hcf;
printf("HCF = %d\n", hcf);
printf("LCM = %d", lcm);
return 0;
}

```

Output:

```

Enter two numbers: 12 18
HCF = 6
LCM = 36

```

8. Perfect Number

Check whether a number is perfect or not.

Example: $6 = 1 + 2 + 3$

Program:

```

#include <stdio.h>

int main()
{
    int num, sum = 0;
    printf("Enter a number: ");
    scanf("%d", &num);
    for(int i = 1; i < num; i++)
    {
        if(num % i == 0)
        {
            sum = sum + i;
        }
    }
    if(sum == num)
    {
        printf("Perfect Number");
    }
    else
    {
        printf("Not a Perfect Number");
    }
    return 0;
}

```

Output:

```

Enter a number: 6
Perfect Number

```

9. Decimal to Binary Conversion

Convert decimal number to binary using loop.

Program:

```
#include <stdio.h>

int main()
{
    int num, binary = 0, place = 1, rem;
    printf("Enter decimal number: ");
    scanf("%d", &num);
    while(num != 0)
    {
        rem = num % 2;
        binary = binary + (rem * place);
        place = place * 10;
        num = num / 2;
    }
    printf("Binary number = %d", binary);
    return 0;
}
```

Output:

```
Enter decimal number: 9
Binary number = 1001
```

10. Menu Driven Calculator

Create calculator using

- for,
- switch
- loops.

Features:

- Addition
- Subtraction
- Multiplication
- Division
- Continue option

Program:

```
#include <stdio.h>

int main()
{
    int choice, i;
    float num1, num2;
    for(i = 1; i > 0; i++)
    {
```

```

printf("\n----- Calculator Menu ----- \n");
printf("1. Addition\n");
printf("2. Subtraction\n");
printf("3. Multiplication\n");
printf("4. Division\n");
printf("5. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
if(choice == 5)
{
    printf("Calculator Closed");
    break;
}
printf("Enter two numbers: ");
scanf("%f%f", &num1, &num2);
switch(choice)
{
    case 1: printf("Addition = %.2f", num1 + num2); break;
    case 2: printf("Subtraction = %.2f", num1 - num2); break;
    case 3: printf("Multiplication = %.2f", num1 * num2); break;
    case 4:
        if(num2 != 0)
            printf("Division = %.2f", num1 / num2);
        else
            printf("Division by zero is not possible");
        break;
    default: printf("Invalid Choice");
}
}
return 0;
}

```

Output:

```

----- Calculator Menu -----
1. Addition  2. Subtraction  3. Multiplication  4. Division  5. Exit
Enter your choice: 1
Enter two numbers: 2 3
Addition = 5.00
Enter your choice: 5
Calculator Closed

```

11. Student Result System

Input marks of 5 students using loop and print:

- Total
- Percentage
- Grade

Program:

```

#include <stdio.h>

int main()
{

```

```

int i;
float sub1, sub2, total, percentage;
for(i = 1; i <= 5; i++)
{
    printf("\nStudent %d\n", i);
    printf("Enter marks of Subject 1: ");
    scanf("%f", &sub1);
    printf("Enter marks of Subject 2: ");
    scanf("%f", &sub2);
    total = sub1 + sub2;
    percentage = total / 2;
    printf("Total = %.2f\n", total);
    printf("Percentage = %.2f\n", percentage);
    if(percentage >= 90)        printf("Grade: A\n");
    else if(percentage >= 75)  printf("Grade: B\n");
    else if(percentage >= 50)  printf("Grade: C\n");
    else                        printf("Grade: Fail\n");
}
return 0;
}

```

Output:

```

Student 1
Enter marks of Subject 1: 90  Enter marks of Subject 2: 95
Total = 185.00  Percentage = 92.50  Grade: A
Student 2
Enter marks of Subject 1: 97  Enter marks of Subject 2: 98
Total = 195.00  Percentage = 97.50  Grade: A

```

12. Employee Salary System

Input salary of multiple employees and calculate:

- Bonus
- Tax
- Final Salary

Using loops + conditions.

Program:

```

#include <stdio.h>

int main()
{
    int n, i;
    float salary, bonus, tax, finalSalary;
    printf("Enter number of employees: ");
    scanf("%d", &n);
    for(i = 1; i <= n; i++)
    {
        printf("\nEnter salary of Employee %d: ", i);
        scanf("%f", &salary);
        if(salary >= 50000)
            bonus = salary * 0.20;
    }
}

```

```

        else
            bonus = salary * 0.10;
        if(salary >= 70000)
            tax = salary * 0.15;
        else
            tax = salary * 0.05;
        finalSalary = salary + bonus - tax;
        printf("Bonus = %.2f\n", bonus);
        printf("Tax = %.2f\n", tax);
        printf("Final Salary = %.2f\n", finalSalary);
    }
    return 0;
}

```

Output:

```

Enter number of employees: 2
Enter salary of Employee 1: 700000
Bonus = 140000.00 Tax = 105000.00 Final Salary = 735000.00
Enter salary of Employee 2: 300000
Bonus = 60000.00 Tax = 45000.00 Final Salary = 315000.00

```

13. Online Exam Result System

Input marks of 5 subjects for multiple students.

Using loop:

- Calculate:
 - Total
 - Percentage
 - Grade
- Print Pass/Fail

Program:

```

#include <stdio.h>

int main()
{
    int students, i;
    float s1, s2, s3, s4, s5, total, percentage;
    printf("Enter number of students: ");
    scanf("%d", &students);
    for(i = 1; i <= students; i++)
    {
        printf("\nEnter marks of Student %d\n", i);
        printf("C: "); scanf("%f", &s1);
        printf("Java: "); scanf("%f", &s2);
        printf("DBMS: "); scanf("%f", &s3);
        printf("HTML: "); scanf("%f", &s4);
        printf("React: "); scanf("%f", &s5);
        total = s1 + s2 + s3 + s4 + s5;
        percentage = total / 5;
        printf("\nTotal = %.2f\n", total);
        printf("Percentage = %.2f\n", percentage);
    }
}

```

```

        if(percentage >= 90)      printf("Grade: A\n");
        else if(percentage >= 75) printf("Grade: B\n");
        else if(percentage >= 50) printf("Grade: C\n");
        else                     printf("Grade: Fail\n");
        if(s1<35||s2<35||s3<35||s4<35||s5<35)
            printf("Result: Fail\n");
        else
            printf("Result: Pass\n");
    }
    return 0;
}

```

Output:

```

Enter number of students: 2
Enter marks of Student 1: C:90 Java:92 DBMS:93 HTML:96 React:97
Total = 468.00  Percentage = 93.60  Grade: A  Result: Pass
Enter marks of Student 2: C:92 Java:70 DBMS:79 HTML:99 React:91
Total = 431.00  Percentage = 86.20  Grade: B  Result: Pass

```

14. Mobile Recharge System

A shopkeeper recharges multiple customers.

Recharge Plans:

- Rs.199
- Rs.399
- Rs.599

Using loop:

- Input customer recharge
- Calculate total earnings of shopkeeper

Program:

```

#include <stdio.h>

int main()
{
    int customers, i, plan;
    float totalEarnings = 0;
    printf("Enter number of customers: ");
    scanf("%d", &customers);
    for(i = 1; i <= customers; i++)
    {
        printf("\nCustomer %d\n", i);
        printf("Available Plans: 1.Rs.199  2.Rs.399  3.Rs.599\n");
        printf("Choose recharge plan amount: ");
        scanf("%d", &plan);
        if(plan == 199 || plan == 399 || plan == 599)
        {
            totalEarnings = totalEarnings + plan;
            printf("Recharge Successful\n");
        }
        else

```

```

        {
            printf("Invalid Recharge Plan\n");
        }
    }
    printf("\nTotal Earnings of Shopkeeper = Rs.%.2f", totalEarnings);
    return 0;
}

```

Output:

```

Enter number of customers: 2
Customer 1 - Choose recharge plan amount: 199 - Recharge Successful
Customer 2 - Choose recharge plan amount: 399 - Recharge Successful
Total Earnings of Shopkeeper = Rs.598.00

```

15. Hospital Patient Temperature Monitor

A hospital records temperature of patients.

Using loop:

- Input temperature of 20 patients
- Count:
 - Normal patients
 - Fever patients (>100 degrees F)

Program:

```

#include <stdio.h>

int main()
{
    int i;
    float temp;
    int normal = 0, fever = 0;
    for(i = 1; i <= 20; i++)
    {
        printf("Enter temperature of Patient %d: ", i);
        scanf("%f", &temp);
        if(temp > 100)
            fever++;
        else
            normal++;
    }
    printf("\nNormal Patients = %d\n", normal);
    printf("Fever Patients = %d\n", fever);
    return 0;
}

```

Output:

```

Enter temperature of Patient 1: 35 ... Patient 20: 36
Normal Patients = 15
Fever Patients = 5

```